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Randomness improvement of chaotic maps for image encryption in a wireless communication scheme using PIC-microcontroller via Zigbee channels

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ABSTRACT

Recently, a lot of research has been done in chaotic cryptography field using different kinds of chaotic systems, like chaotic maps, which are being considered as one of the secure and efficient methods to protect confidential information. This article highlights that the main cryptography requirements demand that the new embedded cryptosystems have to be more efficient and secure, it means that they must be faster and offer greater security. For instance, the new cryptosystems require to be compatible with the new telecommunication protocols and, in addition, to be efficient in energy consumption. In this manner, this article introduces a process to improve the randomness of five chaotic maps that are implemented on a PIC-microcontroller. The improved chaotic maps are tested to encrypt digital images in a wireless communication scheme, particularly on a machine-to-machine (M2M) link, via ZigBee channels. We show that function *mod* 255 improves the randomness of the pseudo-random number generators (PRNG), which is verified performing NIST SP 800-22 statistical tests, histograms, phase-plane analysis, entropy, correlation of adjacent pixels, differential attacks, and using digital images of size 256 × 256 and 512 × 512 pixels. A comparative analysis is presented versus related works that also use chaotic encryption and classic algorithms, such as: AES, DES, 3DES and IDEA. The security analysis confirms that the proposed process to improve the randomness of chaotic maps, is appropriate to implement an encryption scheme that is secure and robust against several known attacks and other statistical tests. Finally, it was experimentally verified that this chaotic encryption scheme can be used in practical applications such as M2M and Internet of things (IoT).

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1. Introduction

With the rapid development of internet, multimedia technologies, communication technologies, microcontroller units (MCU) and internet protocols, the number of services and multiple modern electronic applications that carry out the exchange of confidential information through public telecommunications channels using electronic devices or embedded systems (ES) is growing exponentially [1,2]. These developments make necessary the discovery of new methods to guarantee that the information being transmitted is secure against known attacks. For several years, researchers have applied classical encryption schemes, like Data Encryption Standard (DES), Advanced Encryption Standard (AES), Triple Data Encryption Standard (3DES), International Data Encryption Algorithm (IDEA) and Rivest Cipher 4 (RC4), which are based on iterated processes, series of linked mathematical operations and Feistel structure, where several rounds of substitution boxes (S-boxes) and permutation boxes (P-boxes) are programmed with exclusive-or (XOR) or bitwise rotation. However, these schemes are excellent algorithms just for text encryption [3,4], but they are not suitable to encrypt digital images securely and efficiently [5], because of its distinguishing features including bulk data capacity, high redundancy among the raw pixels and strong adjacent pixels correlation [6–8]. These issues make the classical schemes be unsuitable for fast communications or real time encryption [9,10], mainly due to the fact that these ciphers require a large computational time and

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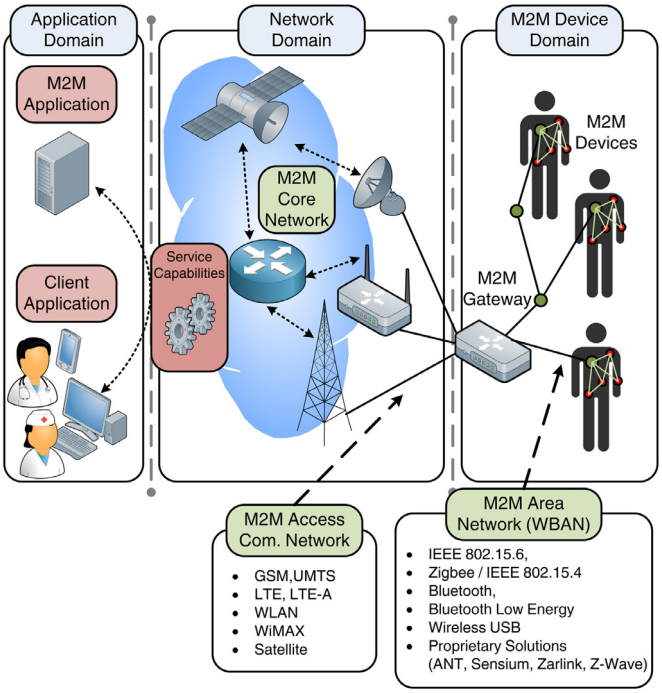

Fig. 1. M2M architecture for wireless connectivity, taken from [34].

high energy consumption [4,11]. Nevertheless, chaotic cryptography is being considered as one of the most secure methods to protect confidential information [12–18], because of its particular properties, such as high sensitivity and dependence to initial conditions, unpredictable behavior, ergodicity, randomness, topology complexity and excellent adaptability to secure communication, among others [19–23]. Besides, some research efforts deal with the combination of AES/DES with chaotic algorithms [24–28], even combining up to four encryption algorithms (DNA-RSA-DES-Chebyshev) [29] with the aim of improving security. However, the execution of these algorithms is too much computational load for an embedded system or MCU. Although, Microchip manufacturer offers a hardware crypto engine on the PIC24F and PIC 32MZ devices for cryptographic functions with AES, DES and 3DES algorithms, this solution is provided by means of a single library for typical applications, such as: web access, secure XML transactions, virtual private networks (VPN) and secure transfer of stored calibration data. As one can infer, a better solution is needed to cope with the modern applications devoted to the Internet of Things (IoT), which expects that by the year 2020 more than 50,000 million devices with digital communication technology will be connected to Internet using embedded systems [30–32]. The IoT enables exchange of information in a wide variety of applications such as smart buildings, smart health, smart transport, Industry 4.0, and so on. As billions of connected things communicate with each other and can exchange sensitive information that may be leaked [33]. The IoT can connect people to people, people to machine or machine to machine (M2M), for example: Fig. 1 depicts the architecture of a (M2M) system [34]. One can see that a better method must be required to guarantee security in the IoT and protect the information against several attacks. Also it must be required to perform cryptanalysis to cryptosystems as reported in the literature [35–39]. The problem is more complex because nowadays intruders and hackers continue to become more sophisticated, so that encryption technology must evolve as well. This problem makes necessary the development of new embedded cryptosystems with greater complexity (enhanced security) and efficiency. In this manner, main contribution of this work focuses on adding security in M2M Area Network by performing chaotic encryption on a wireless communication scheme with Zigbee/IEEE 802.15.4 protocol.

Designing secure M2M systems requires the ability to design fast and secure cryptographic modules, in which the main component is the pseudorandom number generator (PRNG), which can be built using chaotic maps.

From our knowledge and based on the reviewed literature, few works report the use of embedded systems for the chaotic encryption of information and its application for M2M and IoT [40–42]. In this manner, the aim of this paper is the introduction of a secure algorithm to improve the dynamics of five chaotic maps implemented on PIC-microcontrollers, with application to encrypt digital images as a confidential information for secure wireless communications on M2M systems and analyze its feasibility to be conditioned for IoT applications.

The rest of this paper is organized as follows: Section 2 presents the proposed chaotic maps to encrypt digital images. Section 3 describes the proposed scheme for M2M solution. Section 4 provides the results of security analysis, performance and comparison with related work. Conclusions are presented in Section 5.

2. Chaotic maps to encrypt digital images

Chaotic maps are attractive to be implemented in encryption algorithms, due to their discrete nature and complexity in their dynamic behavior. A great variety of chaotic models implemented in embedded systems are reported in the literature, see for example [43–58].

In this paper we present the implementation of five chaotic maps in the microcontroller PIC 16F877A, such as: Hénon [59], Tinkerbell [60,61], Chen [62], Rössler [63] and Logistic 2D [64] maps.

The Eq. (1) represents the Hénon Map [59]. The dynamics depend on two parameters $a = 1.4$, $b = 0.3$, with initial conditions $x_0 = 0.10$ and $y_0 = 0.15$ to generate a chaotic behavior,

$$\begin{aligned} x_{n+1} &= y_n + 1 - ax_n^2, \\ y_{n+1} &= bx_n. \end{aligned} \tag{1}$$

The Tinkerbell map [60,61] exhibits an interesting dynamic behavior represented by Eqs. (2).

$$\begin{aligned} x_{n+1} &= x_n^2 - y_n^2 + ax_n + by_n, \\ y_{n+1} &= 2x_ny_n + cx_n + dy_n, \end{aligned} \tag{2}$$

where parameters $a = 0.9, b = -0.6013, c = 2$, $d = 0.5$, and with initial conditions $x_0 = -0.72$ and $y_0 = -0.64$.

The Chen hypercaotic map [62] is described by the Eqs.in differences (3),

$$\begin{aligned} x_{n+1} &= 1 - a(x_n^2 + y_n^2), \\ y_{n+1} &= 2 - abx_ny_n, \end{aligned} \tag{3}$$

where parameters $a = 1.95$ and $b = 1$, and initial conditions $x_0 = 0.025$ and $y_0 = 0.025$.

The Rössler hypercaotic map [63], is composed by Eqs. (4) that depends on the parameters $\alpha, \beta, \gamma, \delta, \zeta, \eta$ y θ.

$$\begin{aligned} x_1(k+1) &= \alpha x_1(k)(1 - x_1(k)) - \beta(x_3(k) + \gamma)(1 - 2x_2(k)), \\ x_2(k+1) &= \delta x_2(k)(1 - x_2(k)) + \zeta x_3(k), \\ x_3(k+1) &= \eta((x_3(k) + \gamma)(1 - 2x_2(k)) - 1)(1 - \theta x_1(k)), \end{aligned} \tag{4}$$

where the parameters $\alpha = 3.8$, $\beta = 0.05$, $\gamma = 0.35$, $\delta = 3.78$, $\zeta = 0.2$, $\eta = 0.1$, $\theta = 1.9$, with initial conditions $x_1(0) = 0.10$, $x_2(0) = 0.15$ and $x_3(0) = 0.01$.

The Logistic 2D map described by Eq. (5), is a model applied to the analysis of population growth and was published by Pierre Verhulst [65]. Its behavior in chaotic regime arises from nonlinear

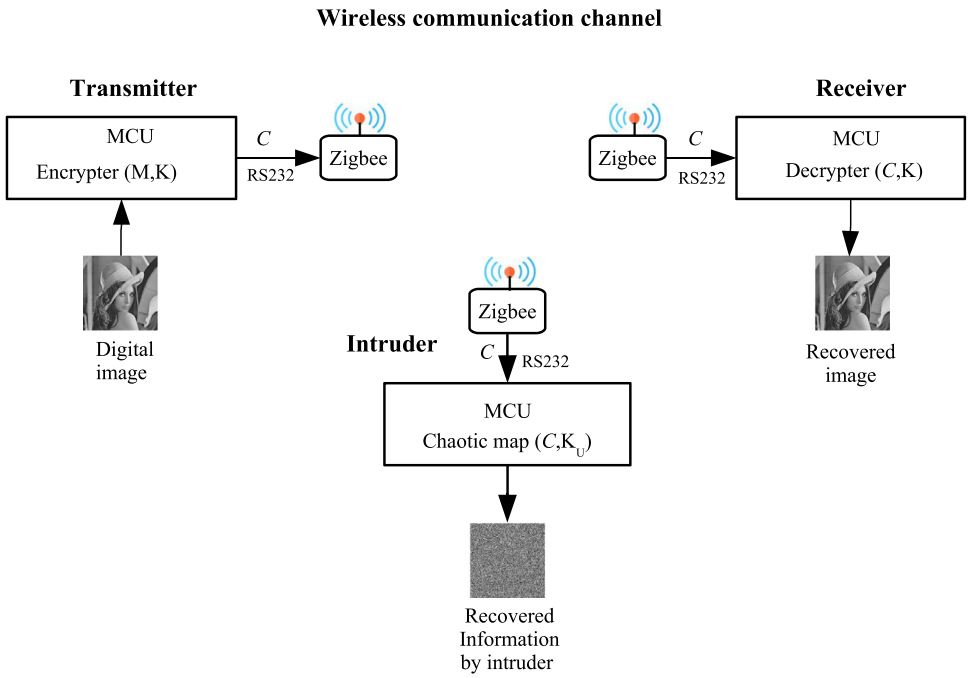

Fig. 2. Scheme of chaotic encryption on a wireless communication system by using Zigbee standard.

dynamics equations, and it has been used for encrypting images with DNA coding [66].

$$x_{n+1} = \mu_1 x_n(1-x_n) + \gamma_1 y_n^2,$$
$$y_{n+1} = \mu_2 y_n(1-y_n) + \gamma_2(x_n^2 + x_n y_n), \quad (5)$$

where parameters $2.75 < \mu_1 \leq 3.4$, $2.75 < \mu_2 \leq 3.45$, $0.15 < \gamma_1 \leq 0.21$, $0.13 < \gamma_1 \leq 0.15$, with initial conditions $x_0 = 0.1$ and $y_0 = 0.3$.

3. Proposed scheme for M2M applications using wireless communications.

The modern systems on a chip (SoC), typically include a microprocessor or microcontroller as a central processing unit in real time, a RAM and FLASH memory, some of the communication ports, such as RS-232, SPI, I2C, USB, Bluetooth, WiFi, Ethernet, GPRS, GSM, among others. They also embed an oscillator clock for synchronization of its internal operations and an energy subsystem. For its proper operation and information management is recommended to use the firmware of communication protocol for sending information through a wireless communication channel such as Zigbee, WiFi, or any other. Fig. 2 illustrates the functional blocks of the proposed solution for an application using wireless communication links based on M2M architecture. It can be observed that for the implementation of this scheme, a central device a MCU PIC 16F877A, which is coded in C language, is used to execute a chaotic map as a pseudorandom number generator (PRNG). Then the MCU encrypts the digital image through the XOR operation using the method reported in [45]. The output (*C-Cryptogram*) of the embedded cryptosystem is send via the RS-232 port to be transmitted using an RF transmitter, such as a ZigBee or WiFi module. In the receiver, the inverse process to encryption is performed. Three MCU PIC16F877A were linked by means of the Zigbee standard, which represent the transmitter, receiver and intruder/hacker in the M2M scheme. The setup of the Zigbee transmitters was done using the XCTU software, which is a free, multi-platform application compatible with Windows, Mac OS and GNU/Linux. The parameters configured for

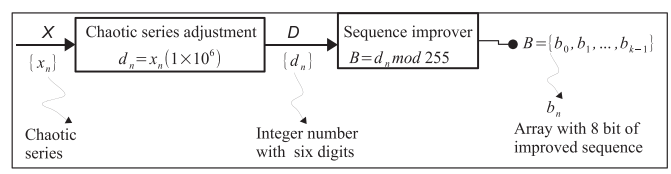

Fig. 3. Algorithm to improve chaotic series for encryption of digital images.

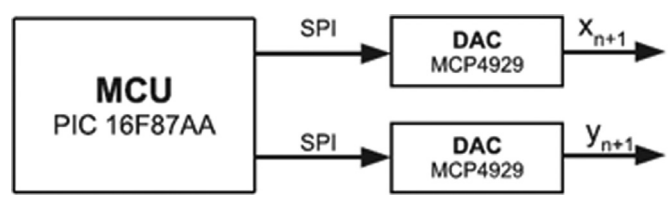

Fig. 4. Block diagram of the chaotic signal generator, also known as Pseudo Random Number Generator (PRNG).

wireless communication are: $Baud = 115200$, $FlowControl = NONE$, $DataBits = 8$, $Parity = NONE$, $StopBits = 1$. For the modem setup, the parameters to be configured are: PAN ID, Destination Address High, Destination Address Low (Serial Number Low). In addition, an intruder/hacker (unauthorized user) can be linked to the Zigbee channel, which tries to extract confidential information. When a digital image is to be encrypted, it is received via the RS-232/USB serial port of the MCU. On the side of the receiver, the inverse process to the encryption is performed and the confidential information is recovered, which is sent via the RS-232/USB port to a personal computer for deployment or storage.

Fig. 3 sketches the modifications to the encryption process already published in [45], where the chaotic sequences are obtained using 32-bit floating point arithmetic according to the IEEE 754 standard [67], to be mapped in 8-bit binary sequences. In the proposed encryption process the chaotic series is multiplied by a constant 1×10^6 to get 32-bit integers, then we propose using *mod* 255 function to convert them to a binary sequences with 8 bit format. This binary sequence is used in this work to encrypt the digi-

Table 1
Comparison of proportion results according to the 16 statistical tests of NIST [68] applied to the five chaotic maps.

NIST SP 800-22 Statistical test	Proportion of success of different chaotic maps.										
	Tinkerbell [60,61]		Rössler [63]			Logistic 2D [64]		Chen [62]		Hénon [59]	
	x_{n+1}	y_{n+1}	x_{n+1}	y_{n+1}	z_{n+1}	x_{n+1}	y_{n+1}	x_{n+1}	y_{n+1}	x_{n+1}	y_{n+1}
Frequency	0.98	0.98	1.00	0.98	1.00	0.99	0.99	0.98	0.99	0.99	0.99
Block Frequency	0.99	0.99	1.00	1.00	1.00	0.99	1.00	1.00	0.99	1.00	1.00
Cumulative sums-Forward	0.99	0.99	1.00	0.98	0.98	0.99	0.99	0.99	0.99	0.99	1.00
Cumulative sums-Reverse	0.97	0.99	1.00	0.98	1.00	0.99	0.98	0.98	1.00	0.99	1.00
Runs	0.99	0.97	0.97	1.00	0.96	0.96	0.99	0.97	0.98	0.90	0.95
Long Runs of Ones	0.95	0.94	0.94	0.94	0.95	0.96	0.94	0.96	0.96	0.94	0.92
Rank	1.00	0.98	0.99	0.99	0.98	1.00	0.98	1.00	0.99	1.00	0.98
Spectral DFT	1.00	0.99	0.98	1.00	1.00	0.99	1.00	0.97	0.99	0.99	0.99
Non-overlapping Templates	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
Overlapping Templates	0.97	0.96	0.98	0.98	0.97	0.99	0.97	0.97	0.98	0.96	0.97
Universal	0.97	0.98	0.98	0.98	0.98	0.98	1.00	1.00	0.98	0.97	1.00
Approximate Entropy	0.95	0.98	0.98	0.96	0.96	0.99	0.99	0.99	0.95	0.96	0.96
Random Excursions	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.98
Random Excursions Variant	0.98	0.99	0.98	0.98	0.98	0.99	0.98	0.98	0.98	0.98	0.98
Linear Complexity	0.99	1.00	0.98	0.98	0.99	1.00	1.00	0.99	0.98	0.98	0.99
Serial ($2m\nabla\Psi$)	0.98	0.99	0.99	0.99	0.99	0.99	0.98	0.99	0.99	0.98	0.96
Average	0.98	0.9812	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.97	0.97

tal images by means of the XOR operation. It is shown that this improves the randomness of chaotic signal, as presented in Section 4.

Algorithm 1 describes the main processes to be executed by the MCU to encrypt and transmit the confidential information. In the receiver, the inverse encryption is performed.

Algorithm 1 Encryption algorithm.

```
#include <16F877A.h>
#include <includes.h> // Include Software UART
#fuses XT,NOWDT                    // Setup Quartz Crystal and disable Watchdog
#use delay(clock = 4000000)        // Clock at 4 MHz
#use rs232(baud = 115200, xmit = PINC6, rcv = PINC7, bits = 8, parity = N) //Setup hardware RS-232
Initializes Software UART
Initializes conditions and variables of chaotic map
Initializes parameters
While (true)
Compute chaotic series
Adjust chaotic series by (1 × 10^6) to obtain x_adj
Improve chaotic data and convert to binary by using mod(x_adj, 255)
Read digital image
Performs XOR operation between chaotic binary data and each pixel of digital image
Send encryption result by Zigbee channel
End While
End program
```

3.1. Pseudo random number generator based on MCU PIC16F877A

Fig. 4 shows the block scheme of the circuit based on MCU PIC16F877A, which is proposed to generate chaotic sequences. The MCU is coded in C language to execute a chaotic map for the generation of pseudo-random numbers (PRNG). In order to observe and verify the chaotic behavior of the generated chaotic sequences, these series go through a process of digital conversion to analog through the DAC MCP4921 or DAC0800.

4. Security analysis

To test the robustness, statistical properties and security of the proposed embedded cryptosystem, the following security analysis were performed: Statistical tests of NIST for cryptographic modules according to SP 800-22, statistical histograms, phase space analysis, key space for the purpose of verify resistance to brute force attacks, correlation analysis of adjacent pixels, correlation between original and encrypted image, analysis of information entropy, NPCR and UACI differential attacks. The quality evaluation of the randomness is carried out to verify the satisfactory security of the new proposed chaos-based cryptosystem.

4.1. Statistical test NIST SP 800-22

Table 1 lists the comparative results obtained with NIST SP 800-22 statistical test suit [68] applied to the binary sequences generated with the different chaotic maps used in this paper by using simple precision according to IEEE 754 Stantard for floating point arithmetic [67]. All test were performed using 100 sequences of stream length = 1,000,000 bit. For these tests, the decision rule at the 1% level was used, i.e., $\alpha = 0.01$, therefore, if the computed P-value is $\geq \alpha$, then one can conclude that the sequence is random with a confidence of 99%. Otherwise, one can conclude that the sequence is non-random with a confidence of 99%. Besides, if the proportion of success is greater than 0.96, it can be concluded that the sequences pass the NIST tests, i.e. those are random sequences. In Table 1, it can be observed that most of the NIST tests pass successfully, except the Long Runs of Ones test, however, as 15 tests pass successfully, of 16 tests, it can be concluded that the majority pass the NIST test.

4.2. Statistical histograms

Fig. 5 depicts the histograms of the Logistic 2D [64], in 5(a) and (b) shows the x_1 and x_2 states respectively. In Fig. 5(c) and 5(d) depicts the improved x_1 and x_2 states, it can be observed that the distribution of histograms is close to uniform, which verifies that the statistics of the chaotic dynamics are improved using the procedure described in Fig. 3. It is important to mention, that the same improvement of the chaotic dynamics were obtained with the Hénon [59], Tinkerbell [60,61], Chen [62] and Rössler [63] maps.

4.3. Phase space analysis

Fig. 6 shows the phase plane representation x_1 vs x_2 of the Logistic 2D chaotic map. Fig. 6(a) shows the original attractor and Fig. 6(b) depicts the improved attractor, where the random scattering of points throughout its phase plane is observed. Again, it

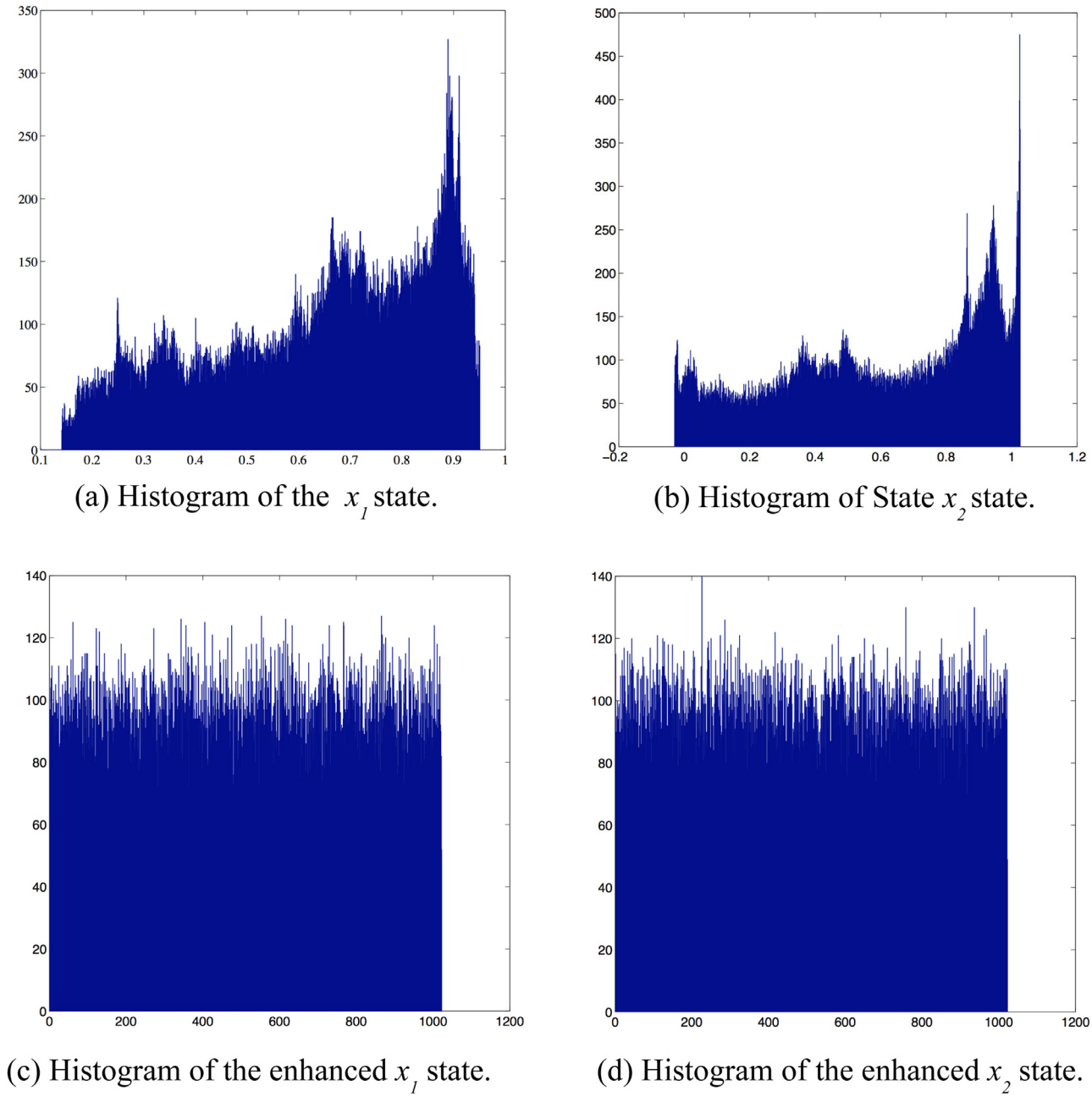

Fig. 5. Histograms of the generated chaotic series with the Logistic 2D map [64]. (a) x_1 state, (b) x_2 state, (c) Improved x_1 state, (d) Improved x_2 state.

is confirmed that the chaotic series are improved. The same results were obtained with the Hénon [59], Tinkerbell [60,61], Chen [62] and Rössler [63] maps.

4.4. Key space

The key space is the total number of different keys that can be used in the encrypted or decrypted process [69]. The authors in [70] report that a cryptosystem to be effective and secure, the key space must be large enough to make unfeasible the brute force attack. The key of the proposed cryptosystem consists of two parts: (i) The initial conditions, and (ii) the parameters of the chaotic map. Then, the secret key or seed of a PRNG must have more than 2^{100} possible secret keys according to [71] to resist an exhaustive attack, therefore, it is considered secure for most cryptographic applications in terms of the speed of current computers. In this paper, the five chaotic maps were implemented by using floating-point arithmetic with the simple and double precision format according to the IEEE 754 Standard [67], i.e., the chaotic sequences were temporarily stored in a 32-bit and 64-bit memory registers.

Table 2 lists the key space obtained by the proposed embedded cryptosystem using Hénon [59], Tinkerbell [60,61], Chen [62], Rössler [63] and Logistic 2D [64] chaotic maps. It can be observed that all chaotic maps meet the minimum key space criteria according to [71]. The size of the key space is directly proportional to the amount of parameters and initial conditions of the chaotic map, for this reason Rössler's hypercaotic map [63] has more key space. Hence, the Kerckhoffs's principles are met, which says, the security of the cryptosystem must rest on the security of the key, being supposed to know the rest of the parameters of the cryptosystem [72–74]. Besides, in Table 2, it can be observed that the key space of the Tinkerbell [60,61], Rössler [63] and Logistic 2D [64] maps is

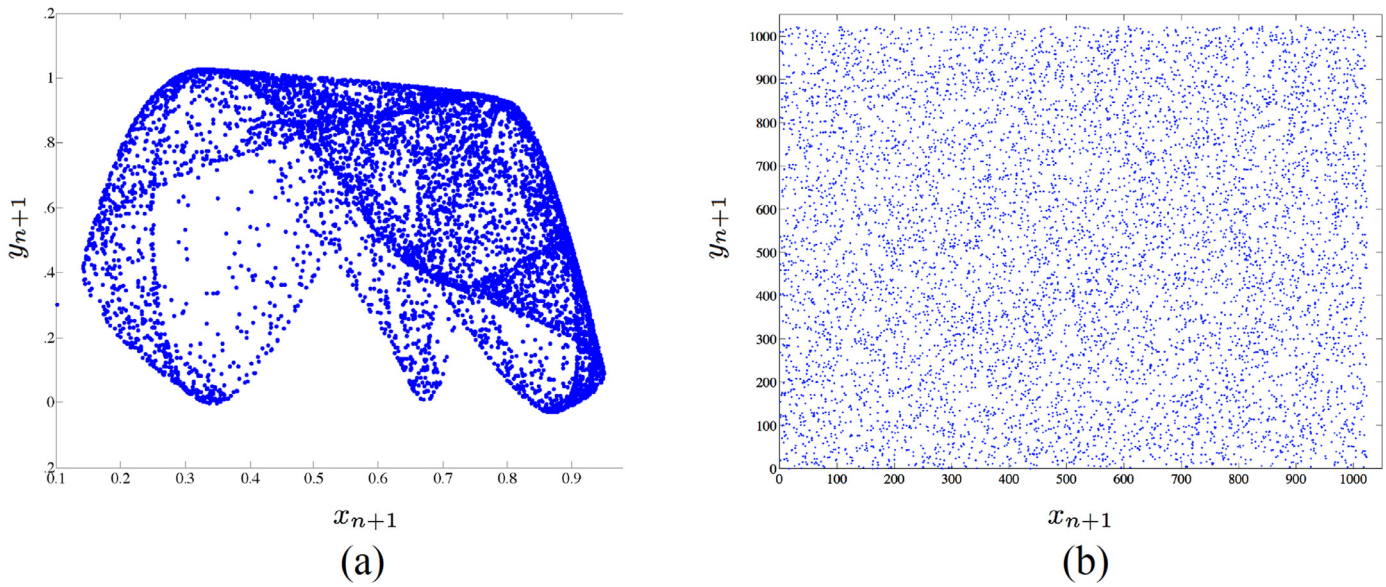

Fig. 6. Phase plane x_1 vs x_2 of the Logistic 2D chaotic map [64]. (a) Original, and (b) Improved attractor.

Table 2
Key space of different chaotic maps implemented by using floating-point arithmetic with simple and double precision format according to the IEEE 754 Standard [67].

Key space		
Chaotic map	Simple precision	Double precision
Hénon [59]	2^{128}	2^{256}
Tinkerbell [60,61]	2^{192}	2^{384}
Chen [62]	2^{128}	2^{256}
Rössler [63]	2^{320}	2^{640}
Logistic 2D [64]	2^{192}	2^{384}
Related work		
Logistic 1D [3]	2^{64}	2^{128}
PELM [75]	2^{64}	2^{128}
Multi modal [8]	2^{80}	2^{159}
Tent map [40]	2^{128}	2^{255}
Novel 3D [22]	2^{95}	2^{189}
Chen [76]	2^{128}	2^{256}
5D Hyperchaotic [77]	2^{150}	2^{299}
Logistic 1D [78]	2^{100}	2^{199}
S-Box & New chaotic system [23]	2^{154}	2^{308}
AES Algorithm [23]	2^{128}, 2^{192}, 2^{256}	2^{128}, 2^{192}, 2^{256}
AES Wavelet & Logistic 1D [26]	2^{128}	2^{256}

greater or equal than all related work that involve chaotic systems as well as classical algorithms.

4.5. Results encrypting digital images

4.5.1. Histograms analaysis

To perform the statistical analysis test of digital image histogram, the well-known image of Lena 512 × 512 in grayscale was used. The x_1 state of Rössler hypercaotic map was selected to perform the information encryption and present its histogram analysis. Fig. 7(a) shows the original image (OI) used to test the proposed encryption algorithm. Fig. 7(d) depicts the corresponding histogram of Lena 512 × 512. Fig. 7(b) shows the criptogram of Lena 512 × 512, it can be observed that it is an unintelligible image by intruders or hackers. Fig. 7(e) depicts the corresponding histogram of encrypted image 512 × 512, it can be observed that the information is distributed over the entire range from 0 to 255 of the grayscale level, i.e., it has a uniform distribution, in the form of random data, therefore, we can say that the proposed cryptosystem is robust against statistical attacks. Fig. 7(c) shows the recovered image of Lena 512 × 512, it can be observed that it is the same as the original image, therefore it is verified that there is no loss of information in the process of encryption and wireless communication. Finally, Fig. 7(f) shows the histogram of the recovered image, from which, it can be seen that it is equal to the histogram of the original image shown in Fig. 7(d), therefore, it is confirmed that there is no loss of confidential information. Regarding the AES algorithm and according to the results of histograms reported in [23], it could be observed that they are not completely uniform across the entire range of the gray scale (0-255), meaning that they have some weakness against statistical attacks [5,69].

4.5.2. Correlation analysis of adjacent pixels

In 1948, Shannon proposed two techniques based on the design of encrypters, the *diffusion* and *confusion*, these two properties above can be demonstrated by a test correlation of adjacent pixels in the encrypted image [69]. It was examined the correlation between two adjacent pixels horizontally, vertically and diagonally. To do this, we randomly selected 5000 pairs of pixels (x_i, y_i) of the digital image under analysis (original or encrypted), and is generated by scattering graphics with these pairs of adjacent pixels, i.e., the pixel is plotted x_i vs y_i. Then their corresponding correlation coefficients (r_{xy}) are calculated by using the next expression [69],

$$r_{xy} = \frac{cov(x,y)}{\sqrt{D(x)}\sqrt{D(y)}}, \tag{6}$$

with

$$cov(x,y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - E(x))(y_i - E(y)), \tag{7}$$

where $cov(x,y)$ is the covariance, $D(x)$ is the variance, x and y denote the scale values of gray level in the digital image under analysis. For this numerical case, the following discrete forms were used:

$$E(x) = \frac{1}{N}\sum_{i=1}^{N}x_i, \tag{8}$$

$$D(x) = \frac{1}{N}\sum_{i=1}^{N}(x_i - E(x))) \tag{9}$$

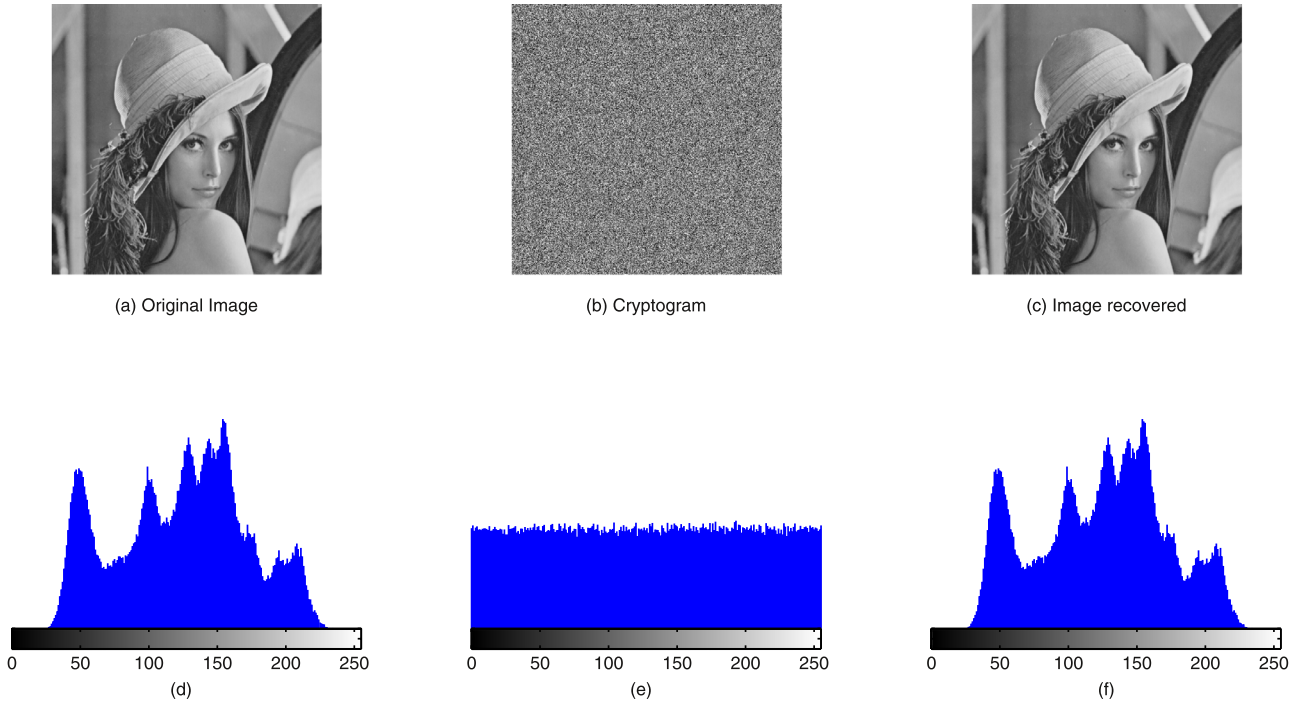

Fig. 7. Digital image encrypted with x_1 state of improved Rössler's hypercaotic map [63]. (a) Original image (OI) Lena 512 × 512. (b) Encrypted image. (c) Recovered image. (d) Histogram of the original image. (e) Histogram of the criptogram. (f) Histogram of the recovered image.

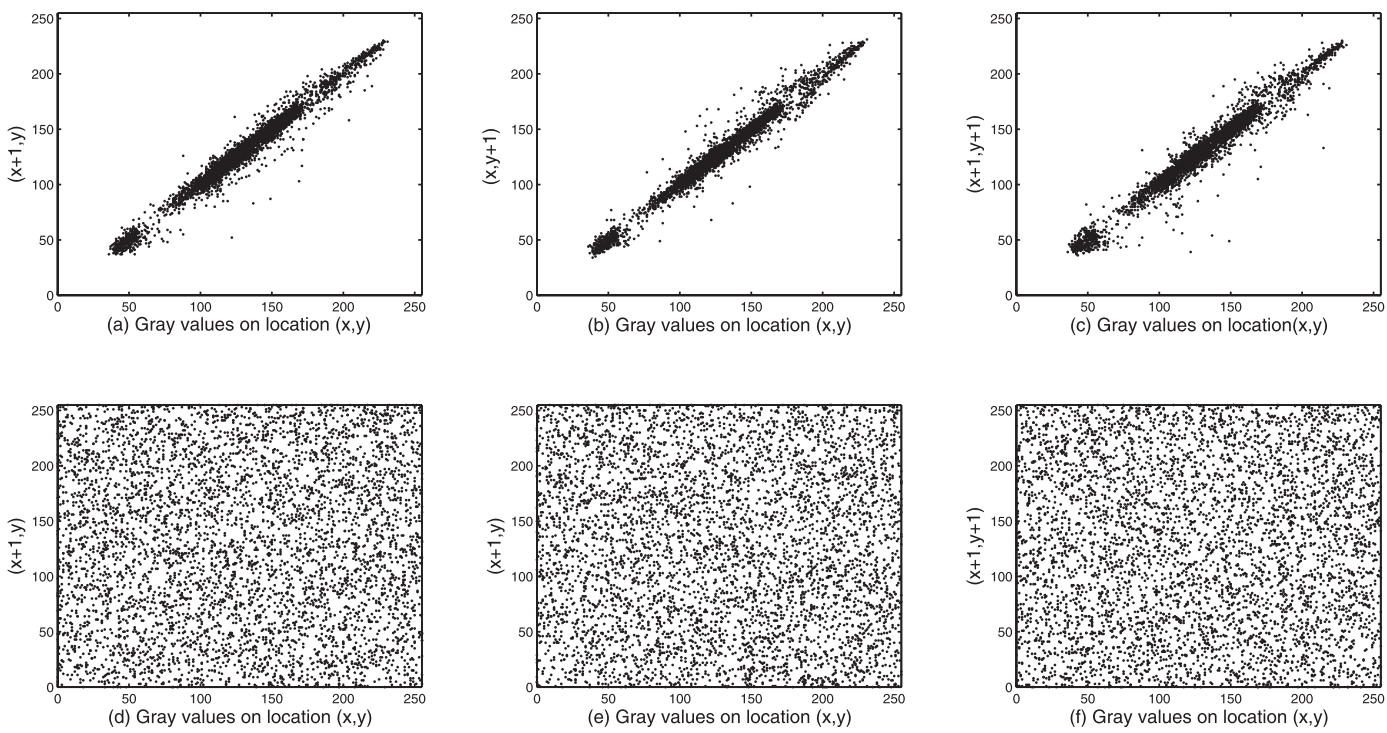

Fig. 8. Correlation of adjacent pixels in the original image Lena 512 × 512 pixels and in the ciphered image: (a) Horizontally adjacent pixels in original image (b) Vertically adjacent pixels in original image. (c) Diagonally adjacent pixels in original image. (d) Horizontally adjacent pixels in encrypted image. (e) Vertically adjacent pixels in encrypted image. (f) Diagonally adjacent pixels in encrypted image.

where $E(x)$ is the average gray levels of pixels.

Fig. 8 depicts the correlation distribution of 5000 pairs of two adjacent pixels in horizontal (a), vertical (b) and diagonal (c) direction of the original digital image Lena 512 × 512 and its corresponding cryptogram obtained with x_1 state of improved Rössler's hypercaotic map [63]. It is well-known that adjacent pixels of an original image are highly correlated, this means that the value of a pixel and its adjacent pixel (horizontal, vertical or diagonal) have a similar value, it can be observed in Fig. 8(a), (b) and –(c). Regarding the pixel correlation (Horizontal, Vertical and Diagonal) of the encrypted image, in Fig. 8(d), (e) and –(f), it can be observed that the correlation of adjacent pixels are com-

Table 3
Correlation coefficients (r_{xy}) of adjacent pixels (horizontal, vertical and diagonal) of the original versus encrypted image.

Chaotic maps	Correlation coefficients (r_{xy}) of adjacent pixels of the image under test			
	OI 256 × 256	Crypt. 256 × 256	OI 512 × 512	Crypt. 512 × 512
Hénon [59]	0.9242	0.0201	0.9855	-0.0021
	0.9529	0.0160	0.9866	0.0203
	0.8936	-0.0165	0.9789	0.0081
Tinkerbell [60,61]	0.9242	0.0033	0.9855	- 3.2225×10^{-04}
	0.9529	-0.0083	0.9866	-6.3715×10^{-04}
	0.8936	-0.0141	0.9789	-0.0071
Chen [62]	0.9242	-0.0246	0.9855	0.0081
	0.9529	-0.0146	0.9866	-0.0070
	0.8936	0.0105	0.9789	-0.0070
Rössler [63]	0.9242	5.4385×10^{-04}	0.9855	0.0039
	0.9529	-0.0122	0.9866	-0.0382
	0.8936	-0.0287	0.9789	0.0108
Logistic 2D [64]	0.9242	0.0079	0.9855	0.0079
	0.9529	0.0011	0.9866	0.0011
	0.8936	0.0369	0.9789	0.0369
Related work				
Multi modal [8]	0.9852	0.0226	N/A	N/A
	0.9658	0.0022	N/A	N/A
	0.9506	-0.0081	N/A	N/A
Chen [76]	0.9488	0.0005	N/A	N/A
	0.9477	0.0005	N/A	N/A
	0.9047	0.0001	N/A	N/A
5D Hyperchaotic [77]	0.9705	-0.0052	N/A	N/A
	0.9414	0.0031	N/A	N/A
	0.9136	-0.0003	N/A	N/A
S-Box & Chaotic syst. [23]	N/A	0.5210	N/A	N/A
	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A
AES [23]	N/A	0.5177	N/A	N/A
	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A
AES & Log. 1D [26]	N/A	N/A	0.9718	0.1394
	N/A	N/A	0.9881	0.0339
	N/A	N/A	0.9749	0.0755
(DNA-RSA-DES-Chebyshev) [29]	N/A	-0.0400	N/A	N/A
	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A

Table 4
Correlation coefficient (r) between original and encrypted image.

Chaotic map	Lena 256 × 256	Lena 512 × 512
Hénon [59]	-0.0054	-0.0011
Tinkerbell [60,61]	0.0027	0.0017
Chen [62]	0.0020	9.4581×10^{-04}
Rössler [63]	-0.0068	-0.0042
Logistic 2D [64]	-1.4370×-04	1.7092×10^{-04}

pletely different, so the dispersion can be observed throughout the plane (x, y).

Table 3 shows the correlation coefficients r_{xy} calculated by using equation (6), it can be observed that the correlation coefficients of the original image are close to 1, which means that there is similarity between the adjacent pixels. With respect to encrypted images, the ideal correlation coefficient is that r_{xy} is equal to zero, it can be observed that in most cases the correlation coefficients are close to zero, however, the most separated from zero were the classical encryption algorithms, such as S-box, AES and DNA-RSA-DES-Chebyshev [23,26,29], which confirms that they have some weakness when are used for image encryption.

Table 4 shows the correlation coefficients r obtained between the original image and the encrypted image. It can be seen that in all cases, the correlation coefficient r is very close to zero, which indicates that there is no similarity between the original and the encrypted image, which confirms that they are completely different. Also, it is verified that the proposed method works for any image size.

4.5.3. Analysis of information entropy

In the works reported by Shannon, the mathematical foundations of the theory of information applied to communication and data storage were proposed. The entropy of the information is a criterion that measures the randomness of the data [46]. It can also be used to evaluate the security of the encryption. Eq. (10) is used to calculate the entropy $H(s)$, of a source (s), [46]

$$H(s) = \sum_{i=0}^{2^N-1} P(s_i) \cdot Log_2\left(\frac{1}{P(s_i)}\right) \text{ bit}, \tag{10}$$

where $P(s_i)$ represents the probability of the s_i symbol. For a purely random source that is emitting 2^N symbols with the same probability after evaluating (10), the entropy $H(s) = N$, in this case, for images with completely random pixels with 8-bit format, its ideal entropy is $H(s) = 8$ bit. When digital images are encrypted, ideally their entropy must be 8. When a cryptographic system emits symbols (cryptograms) with entropy less than 8, this encrypter has a certain degree of predictability, so that its security is put at risk [46]. Table 5 shows the entropy results obtained with the five chaotic maps used in this paper and comparison versus other related works that also report results using the Lena image with 8-bit format. In Table 5, it can be observed that entropy is not affected by the size of the image and the results are very similar to those reported in related work. In addition, it can be observed that classical encryption algorithms, such as S-box and AES [23,26],

Table 5
Comparison of entropy results for cryptogram of digital images size 256 × 256 and 512 × 512.

Chaotic map	Size 256 × 256	Size 512 × 512
Hénon [59]	7.9973	7.9992
Tinkerbell [60,61]	7.9969	7.9993
Logistic 2D [64]	7.9971	7.9971
Chen [62]	7.9971	7.9993
Rössler [63]	7.9971	7.9993
Related work		
Logistic 1D [3]	7.9948	7.9972
Multi modal [8]	7.9989	N/A
Novel 3D [22]	N/A	7.9982
Chen [76]	7.9972	N/A
5D Hyperchaotic [77]	7.9973	N/A
Logistic 1D [78]	N/A	7.9991
S-Box & New chaotic syst. [23]	N/A	7.95667
AES Algorithm [23]		7.95912
AES & Logistic 1D [26]	N/A	7.7362 Avg

are farther from the ideal value 8, which means that they have a certain degree of predictability, therefore its security is put at risk [46], this confirms that they were not designed to encrypt images [4,5].

4.5.4. NPCR and UACI differential attacks

To perform an analysis against differential attacks and understand the differences between encrypted images [46], two measures in common are used, NPCR and UACI. These measures are used to test the influence of change of a pixel in the whole encrypted pattern. Let us consider the cryptograms C_1 and C_2 obtained with a tiny difference of 1^{-10} in the encryption key (x_1) of the receiver and using a Lena image 512 × 512 in gray scale. In Table 6, it can be seen that the five chaotic maps proposed in this paper pass the NPCR differential attack test according to the critical values established by [79], even in most cases, is greater than those reported in related work, except the Tinkerbell[60,61] and Rössler[63] map, they are not greater than Chen's map reported in [76]. Also, it can be observed that the classical S-Box and AES encryption algorithms [23,26] pass the NPCR test against differential attack, except the method that combines four encryption algorithms [29].

In Table 7, it can be observed that the five chaotic maps proposed in this paper pass the UACI differential attack test according to the critical values established by [79], therefore, the 100% of the proposed chaotic maps pass the UACI test. However, it can be seen that some related work do not pass the UACI test, such as [8,75] and classical algorithms [23,26,29].

4.5.5. Comparison with classic algorithms

Table 8 highlights that the proposed algorithm has a structure based on chaotic map, and performs stream encryption with symmetric key in a round. Depending on the selected chaotic map, one can have a large key space, from 2^{128} to 2^{640}. It is well known, that stream ciphers have been mathematically simpler, faster and easier to implement in hardware. Therefore, the proposed scheme presents some advantages versus the classic algorithms, mainly it is more secure, as demonstrated by the security analysis. This means that it provides better robustness against different attacks, such as: key space, statistical histogram, correlation of adjacent pixels, information entropy, NPCR and UACI differential attacks.

To perform a good performance comparison of speed of the proposed scheme versus classical algorithms reported in [9], the proposed algorithm was implemented in an AMD A6 4400M APU 2.7 GHz computer with Windows 10 operating system and Python version 3.5.2. Also, the proposed cryptosystem was divided into two stages, the first is the PRNG and the second is the X-OR operation. All speed tests were performed by generating 1,000,000 bits for each chaotic map and measuring the generation time of the pseudorandom numbers, to later convert to speed [45]. Table 9 lists the processing speeds of the PRNGs by using double precision for each chaotic map [59–64] proposed in this paper and a comparison versus classical algorithms [9]. We can observe that the highest speed are achieved using Henon and Chen map obtaining 12.80 Mbit/s, which are greater than the results reported in [9] with 11.64 Mbit/s using AES (192-bit key). The slowest speeds are obtained by using Rössler map obtaining 6.40 Mbit/s, however is similar to DES algorithm [9] with 6.483 Mbit/s. The speed results obtained with the proposed scheme, proves that can be also used for practical applications and for modern telecommunications devices, such as microcontrollers, smart phones, tablets, etc. Since, the proposed scheme was implemented by using the Python programming language, it has the advantage that uses an open source, free license and multi-platform, i. e., it can be implemented in different types of hardware and in different modern operating systems [45], such as Android, iOS, Windows 10, GNU Linux and OS X. Therefore, the efficiency and speed of the cryptographic modules depends on the quantity of arithmetic and logical operations,

Table 6
NPCR test for the cryptogram of digital image 512 × 512 pixels and comparison versus other related work.

Cryptogram size		Critical values of NPCR [79]		
512 × 512		$N_{0.05} = 99.5893\%$,	$N_{0.01} = 99.5810\%$,	$N_{0.001} = 99.5717\%$,
Chaotic map	**NPCR** (%)		**NPCR Test Result**	
Hénon[59]	99.8093	Passed	Passed	Passed
Tinkerbell[60,61]	99.7990	Passed	Passed	Passed
Logistic 2D[64]	99.8138	Passed	Passed	Passed
Chen[62]	99.8104	Passed	Passed	Passed
Rössler[63]	99.7932	Passed	Passed	Passed
Related work				
Logistic 1D [3]	99.6100	Passed	Passed	Passed
PELM [75]	99.5774	Failed	Failed	Passed
Multi modal [8]	99.0000	Failed	Failed	Failed
Tent map [40]	99.6300	Passed	Passed	Passed
Novel 3D [22]	99.6100	Passed	Passed	Passed
Chen [76]	99.8031	Passed	Passed	Passed
5D Hyperchaotic [77]	99.6122	Passed	Passed	Passed
Logistic 1D [78]	99.6100	Passed	Passed	Passed
S-Box & Chaotic syst. [23]	99.6298	Passed	Passed	Passed
AES Algorithm [23]	99.6324	Passed	Passed	Passed
AES & Log. 1D [26]	N/A	N/A	N/A	N/A
(DNA-RSA-DES-Chebyshev) [29]	98.6000	Failed	Failed	Failed

Table 7
UACI test for cryptogram of digital image 512 × 512 pixels and comparison versus related work.

Cryptogram size		Critical values of UACI [79]		
512 × 512		$U^-_{0.05} = 33.3730\%$, $U^+_{0.05} = 33.5541\%$,	$U^-_{0.01} = 33.3445\%$, $U^+_{0.01} = 33.5826\%$,	$U^-_{0.001} = 33.3115\%$, $U^+_{0.001} = 33.6156\%$,
Chaotic map	**UACI** (%)	**UACI Test Result**		
Hénon [59]	33.4805	Passed	Passed	Passed
Tinkerbell [60,61]	33.4524	Passed	Passed	Passed
Logistic 2D [64]	33.5188	Passed	Passed	Passed
Chen [62]	33.5227	Passed	Passed	Passed
Rössler [63]	33.5160	Passed	Passed	Passed
Related work				
Logistic 1D [3]	33.3600	Passed	Passed	Passed
PELM [75]	33.3014	Failed	Failed	Passed
Multi modal [8]	34.8353	Failed	Failed	Failed
Tent map [40]	33.2800	Passed	Passed	Passed
Novel 3D [22]	33.4500	Passed	Passed	Passed
Chen [76]	33.6236	Failed	Failed	Failed
5D Hyperchaotic [77]	33.4573	Passed	Passed	Passed
Logistic 1D [78]	33.4500	Passed	Passed	Passed
S-Box & Chaotic syst. [23]	31.8345	Failed	Failed	Failed
AES Algorithm [23]	31.8723	Failed	Failed	Failed
AES & Log. 1D [26]	N/A	N/A	N/A	N/A
(DNA-RSA-DES-Chebyshev) [29]	33.1000	Failed	Failed	Failed

Table 8
Characteristics comparison of the proposed scheme versus classical algorithms.

Charasteristics	AES	DES	3DES	IDEA	Proposed algorithm
Algorithm Structure	Feistel	Feistel	Feistel	Feistel	Chaotic map
Cipher type	Block	Block	Block	Block	Stream
Key type	Symmetric	Symmetric	Symmetric	Symmetric	Symmetric
Rounds	10, 12, 14	16	48	8	1
Block size [Bit]	128	64	64	64	N/A
Key size [Bit]	128, 192, 256	56	168, 112, 56	128	128, 192, 256 up to 640
S-Boxes	1	8	8	N/A	N/A
Security for images	Vulnerable	Vulnerable	Vulnerable	Vulnerable	Secure

Table 9
Comparison of processing speed (Mbit/s) versus classic algorithms.

Algorithm	CPU Clock (GHz)	Software	Double precision
Proposed with Henon map [45]	2.7	Python 3.5.2	12.8008
Proposed with Tinkerbell map [45]	2.7	Python 3.5.2	9.1491
Proposed with Chen map [45]	2.7	Python 3.5.2	12.8000
Proposed with Rössler map [45]	2.7	Python 3.5.2	6.4000
Proposed with Logistic 2D map [45]	2.7	Python 3.5.2	9.1491
Classical algorithms			
DES [9]	2.8	Crypto+ library	6.4830
AES (192-bit key)[9]	2.8	Crypto+ library	11.6450
AES (256-bit key) [9]	2.8	Crypto+ library	11.3800

permutations, number of rounds, dimensions of the chaotic maps, software and hardware.

5. Conclusions

Five chaotic maps were used as cases of study to propose a simple, efficient and secure algorithm to encrypt digital images in a wireless communication scheme. The proposed encryption system was implemented on a PIC 16F877A microcontroller and the experimental results were tested using M2M links, via Zigbee channel. The experimental results were obtained with two different sizes of digital images. The security analysis confirmed that the proposed encryption scheme using *mod* 255, offers high security and robustness against several known attacks, i.e., it is strong against several statistical tests. It is worth mentioning that the five chaotic maps used in this paper meet the minimum key space criteria, which must be greater than 2^{100}. Therefore, the Kerckhoffs's principles are met. It was shown that the proposed scheme presents some advantages versus the classic algorithms (AES, DES, 3DES, IDEA), mainly it is highlighted that it is more secure, as it was shown in the security analysis, that is, presents better robustness against different attacks, such as: Key space, statistical histogram, correlation of adjacent pixels, information entropy, NPCR and UACI differential attacks. It was proved that it is not necessary to combine several algorithms in the encryption process to have a good security, according to the capacity of the current computers, it is enough to use a chaotic map without compromising information security. The performance and speed of the chaotic encryption algorithms is slightly better than the classic algorithms. Finally, it was experimentally verified that this chaotic encryption scheme can be used in practical applications such as M2M and IoT.

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Declaration of Competing Interest

The authors declare no conflict of interest. The founding sponsors had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript, or in the decision to publish the results.

CRediT authorship contribution statement

E.E. García-Guerrero: Writing - original draft, Conceptualization. **E. Inzunza-González:** Methodology, Software, Formal analysis, Funding acquisition. **O.R. López-Bonilla:** Investigation, Validation, Resources. **J.R. Cárdenas-Valdez:** Visualization, Project administration, Funding acquisition. **E. Tlelo-Cuautle:** Supervision, Writing - review & editing.

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References

[1] Panagiotou P, Sklavos N, Darra E, Zaharakis ID. Cryptographic system for data applications, in the context of internet of things. Microprocess Microsyst 2020;72:102921. doi:10.1016/j.micpro.2019.102921.
[2] Morabito R, Petrolo R, Loscri V, Mitton N. LEGIoT: a lightweight edge gateway for the internet of things. Future Gener Comput Syst 2018;81:1–15. doi:10.1016/j.future.2017.10.011.
[3] Murillo-Escobar MA, Cruz-Hernández C, Abundiz-Pérez F, López-Gutiérrez RM, del Campo OR. A RGB image encryption algorithm based on total plain image. Signal Process 2015;109:119–31. doi:10.1016/j.sigpro.2014.10.033.
[4] Ye G. A block image encryption algorithm based on wave transmission and chaotic systems. Nonlinear Dyn 2014;75:417–27. doi:10.1007/s11071-013-1074-6.
[5] Murillo-Escobar MA, Meranza-Castillón MO, López-Gutiérrez RM, Cruz-Hernández C. Suggested integral analysis for chaos-based image cryptosystems. Entropy 2019;21(8):815. doi:10.3390/e21080815.
[6] Cao C, Sun K, Liu W. A novel bit-level image encryption algorithm based on 2D-LICM hyperchaotic map. Signal Process 2018;143:122–33. doi:10.1016/j.sigpro.2017.08.020.
[7] Xu L, Li Z, Li J, Hua W. A novel bit-level image encryption algorithm based on chaotic maps. Optics Lasers Eng 2016;78:17–25. doi:10.1016/j.optlaseng.2015.09.007.
[8] García-Martínez M, Campos-Cantón E. Pseudo-random bit generator based on multi-modal maps. Nonlinear Dyn 2015;82(4):2119–31. doi:10.1007/s11071-015-2303-y.
[9] Kwok HS, Tang WKS. A fast image encryption system based on chaotic maps with finite precision representation. Chaos Solitons Fractals 2007;32(4):1518–29. doi:10.1016/j.chaos.2005.11.090.
[10] Liu H, Wang X. Color image encryption based on one-time keys and robust chaotic maps. Comput Math Appl 2010;59(10):3320–7. doi:10.1016/j.camwa.2010.03.017.
[11] Pareek NK, Patidar V, Sud KK. Image encryption using chaotic logistic map. Image Vision Comput 2006;24(9):926–34. doi:10.1016/j.imavis.2006.02.021.
[12] Nardo LG, Nepomuceno EG, Arias-Garcia J, Butusov DN. Image encryption using finite-precision error. Chaos Solitons Fractals 2019;123:69–78. doi:10.1016/j.chaos.2019.03.026.
[13] Bahramian A, Nouri A, Baghdadi G, Gharibzadeh S, Towhidkhah F, Jafari S. Introducing a chaotic map with a wide range of long-term memory as a model of patch-clamped ion channels current time series. Chaos Solitons Fractals 2019;126:361–8. doi:10.1016/j.chaos.2019.07.018.
[14] Lai Q, Norouzi B, Liu F. Dynamic analysis, circuit realization, control design and image encryption application of an extended Lü system with coexisting attractors. Chaos Solitons Fractals 2018;114:230–45. doi:10.1016/j.chaos.2018.07.011.
[15] Khan M, Masood F. A novel chaotic image encryption technique based on multiple discrete dynamical maps. Multimed Tools Appl 2019. doi:10.1007/s11042-019-07818-4.
[16] Kocamaz UE, Cicek S, Uyaroglu Y. Secure communication with chaos and electronic circuit design using passivity-based synchronization. J Circuits Syst Comput 2018;27(4). doi:10.1142/S0218126618500573.
[17] Ping P, Wu J, Mao Y, Xu F, Fan J. Design of image cipher using life-like cellular automata and chaotic map. Signal Process 2018;150:233–47. doi:10.1016/j.sigpro.2018.04.018.
[18] Özkaynak F. Brief review on application of nonlinear dynamics in image encryption. Nonlinear Dyn 2018;92(2):305–13. doi:10.1007/s11071-018-4056-x.
[19] Ge B, Luo H. Image encryption application of chaotic sequences incorporating quantum keys. Int J Autom Comput 2019:1–16. doi:10.1007/s11633-019-1173-z.
[20] Tsai C-J, Wang H-C, Wu JL. Three techniques for enhancing chaos-based joint compression and encryption schemes. Entropy 2019;21(40). doi:10.3390/e21010040.
[21] Natiq H, Said MRM, Al-Saidi NMG, Kilicman A. Dynamics and complexity of a new 4D chaotic laser system. Entropy 2019;21(34). doi:10.3390/e21010034.
[22] Sahari ML, Boukemara I. A pseudo-random numbers generator based on a novel 3D chaotic map with an application to color image encryption. Nonlinear Dyn 2018;94(1):723–44. doi:10.1007/s11071-018-4390-z.
[23] Çavuşoğlu U, Kaçar S, Pehlivan I, Zengin A. Secure image encryption algorithm design using a novel chaos based s-box. Chaos Solitons Fractals 2017;95:92–101. doi:10.1016/j.chaos.2016.12.018.
[24] Zhang X, Wang L, Cui G, Niu Y. Entropy-based block scrambling image encryption using DES structure and chaotic systems. Int J Optics 2019;2019:13. doi:10.1155/2019/3594534.
[25] El Batouty AS, Farag HH, Mokhtar MA, El-Badawy EA. New hybrid AES static s-box algorithm using chaotic maps. In: 2019 International Conference on Information Technologies (InfoTech); 2019. p. 1–5. doi:10.1109/InfoTech.2019.8860892.
[26] Shakir HR. An image encryption method based on selective AES coding of wavelet transform and chaotic pixel shuffling. Multimed Tools Appl 2019;78. doi:10.1007/s11042-019-07766-z.
[27] Zhang L, Yu J, Wu Z. Secured chaotic cognitive radio system using advanced encryption standard. In: 2015 IEEE 26th Annual International Symposium on Personal, Indoor, and Mobile Radio Communications (PIMRC); 2015. p. 7–11. doi:10.1109/PIMRC.2015.7343259.
[28] Atteya AM, Madian AH. A hybrid chaos-AES encryption algorithm and its impelmention based on FPGA. In: 2014 IEEE 12th International New Circuits and Systems Conference (NEWCAS); 2014. p. 217–20. doi:10.1109/NEWCAS.2014.6934022.
[29] Shima Ramesh M, Thanikaiselvan V. A novel efficient multiple encryption algorithm for real time images. Int J Electr ComputEng 2020;10(2):1327–36.
[30] Al-Fuqaha A, Guizani M, Mohammadi M, Aledhari M, Ayyash M. Internet of things: a survey on enabling technologies, protocols, and applications. IEEE Commun Surv Tutor 2015;17(4):2347–76. doi:10.1109/COMST.2015.2444095.
[31] Ng ICL, Wakenshaw SYL. The internet-of-things: review and research directions. Int J Res Market 2017;34(1):3–21. doi:10.1016/j.ijresmar.2016.11.003.
[32] Ashton K. That "Internet of Things" thing. RFiD J 2009;22:97–114.
[33] Harbi Y, Aliouat Z, Harous S, Bentaleb A, Refoufi A. A review of security in internet of things. Wirel Pers Commun 2019. doi:10.1007/s11277-019-06405-y.
[34] Kartsakli E, Lalos AS, Antonopoulos A, Tennina S, Renzo MD, Alonso L, et al. A survey on M2M systems for mhealth: a wireless communications perspective. Sensors 2014;14:18009–52. doi:10.3390/s141018009.
[35] Li M, Lu D, Xiang Y, Zhang Y, Ren H. Cryptanalysis and improvement in a chaotic image cipher using two-round permutation and diffusion. Nonlinear Dyn 2019;96(1):31–47. doi:10.1007/s11071-019-04771-7.
[36] Li M, Zhou K, Ren H, Fan H. Cryptanalysis of permutation-diffusion-based lightweight chaotic image encryption scheme using CPA. Appl Sci-Basel 2019;9(3). doi:10.3390/app9030494.
[37] Liu Y, Qin Z, Wu J. Cryptanalysis and enhancement of an image encryption scheme based on bit-plane extraction and multiple chaotic maps. IEEE Access 2019;7:74070–80. doi:10.1109/ACCESS.2019.2916600.
[38] Kim H, Kim D, Yi O, Kim J. Cryptanalysis of hash functions based on blockciphers suitable for IoT service platform security. Multimed Tools Appl 2019;78(3):3107–30. doi:10.1007/s11042-018-5630-4.
[39] Zhu C, Wang G, Sun K. Improved cryptanalysis and enhancements of an image encryption scheme using combined 1d chaotic maps. Entropy 2018;20. doi:10.3390/e20110843.
[40] Palacios-Luengas L, Pichardo-Méndez JL, Díaz-Méndez JA, Rodríguez-Santos F, Vázquez-Medina R. PRNG based on skew Tent map. Arabian J Sci Eng 2018;44(4):3817–30. doi:10.1007/s13369-018-3688-y.
[41] Jridi M, Chapel T, Dorez V, Guénolé LB, Boltlan LA. SoC-based edge computing gateway in the context of the internet of multimedia things: experimental platform. Low Power Electron Appl 2018;8(1):1–18. doi:10.3390/jlpea8010001.
[42] Azzaz MS, Tanougast C, Sadoudi S, Bouridane A. Synchronized hybrid chaotic generators: Application to real-time wireless speech encryption. Commun Nonlinear Sci NumerSimul 2013;18(8):2035–47. doi:10.1016/j.cnsns.2012.12.018.
[43] Castañeda CE, López-Mancilla D, Chiu R, Villafaña Rauda E, Orozco-López O, Casillas-Rodríguez F, et al. Discrete-time neural synchronization between an arduino microcontroller and a compact development system using multiscroll chaotic signals. Chaos Solitons Fractals 2019;119:269–75. doi:10.1016/j.chaos.2018.12.030.
[44] Teh-Lu L, Pei-Yen W, Jun-Juh Y. Design of synchronized large-scale chaos random number generators and its application to secure communication. MDPI Appl Sci 2019;9(185):1–13. doi:10.3390/app9010185.
[45] Flores-Vergara A, Garcia-Guerrero EE, Inzunza-Gonzalez E, Bonilla ORL, Rodriguez-Orozco E, Cardenas-Valdez JR, et al. Implementing a chaotic cryptosystem in a 64-bit embedded system by using multiple-precision arithmetic. Nonlinear Dyn 2019;96(1):497–516. doi:10.1007/s11071-019-04802-3.
[46] Flores-Vergara A, Inzunza-González E, García-Guerrero EE, López-Bonilla OR, Rodríguez-Orozco E, Hernández-Ontiveros JM, Cárdenas-Valdez JR, Tlelo-Cuautle E. Implementing a chaotic cryptosystem by performing parallel computing on embedded systems with multiprocessors. Entropy 2019;21(3):268. doi:10.3390/e21030268.
[47] Helmy M, El-Rabaie E-SM, Eldokany IM, El-Samie FEA. Chaotic encryption with different modes of operation based on Rubik's cube for efficient wire-

less communication. Multimed Tools Appl 2018;77:27337–61. doi:10.1007/s11042-018-5923-7.
[48] Karthikeyan R, Serdar C, Viet-Thanh P, Sajad J, Anitha K. A novel class of chaotic systems with different shapes of equilibrium and microcontroller-based cost-effective design for digital applications. Eur Phys J 2018;133(231):1–11. doi:10.1140/epjp/i2018-12053-7.
[49] Teh-Lu L, Pei-Yen W, Chien P-C, Yi-Chieh L, Liand-Kai W, Jun-Juh Y. Design of high-security USB flash drives based on chaos authentication. Electronics 2018;7(82):1–9. doi:10.3390/electronics7060082.
[50] Li B, Liao X, Jiang Y. A novel image encryption scheme based on improved random number generator and its implementation. Nonlinear Dyn 2018;95(3):1–25. doi:10.1007/s11071-018-4659-2.
[51] Méndez-Ramírez R, Arellano-Delgado A, Cruz-Hernández C, Abundiz-Pérez F, Martínez-Clark R. Chaotic digital cryptosystem using serial peripheral interface protocol and its dsPIC implementation. Front Inf Technol ElectronEng 2018;19(2):165–79. doi:10.1631/FITEE.1601346.
[52] Janakiraman S, Thenmozhi k, Balaguru Rayappan JB, Amirtharajan R. Lightweight chaotic image encryption algorithm for real-time embedded system: Implementation and analysis on 32-bit microcontroller. Microprocess Microsyst 2018;56:1–12. doi:10.1016/j.micpro.2017.10.013.
[53] Garipcan AM, Erdem E. Implementation and performance analysis of true random number generator on FPGA environment by using non-periodic chaotic signals obtained from chaotic maps. Arabian J Sci Eng 2019:1–15. doi:10.1007/s13369-019-04027-x.
[54] Guillen-Fernandez O, Melendez-Cano A, Tlelo-Cuautle E, Nuñez Perez JC, Rangel-Magdaleno JJ. On the synchronization techniques of chaotic oscillators and their FPGA-based implementation for secure image transmission. PLoS ONE 2019;14(2):1–34. doi:10.1371/journal.pone.0209618.
[55] Karimov T, Butusov D, Andreev V, Karimov A, Tutueva A. Accurate synchronization of digital and analog chaotic systems by parameters re-identification. Electronics 2018;7(123):1–10. doi:10.3390/electronics7070123.
[56] Rezk AA, Madian AH, Radwan AG, Soliman AM. Reconfigurable chaotic pseudo random number generator based on FPGA. AEU - Int J ElectronCommun 2019;98:174–80. doi:10.1016/j.aeue.2018.10.024.
[57] Rodríguez-Orozco E, García-Guerrero EE, Inzunza-Gonzalez E, López-Bonilla OR, Flores-Vergara A, Cárdenas-Valdez JR, Tlelo-Cuautle E. FPGA-based chaotic cryptosystem by using voice recognition as access key. Electronics 2018;7(12). doi:10.3390/electronics7120414.
[58] Shuai C, Bing L, Cenjun Z. FPGA implementation of SRAM PUFs based cryptographically secure pseudo-random number generator. Microprocess Microsyst 2018;59:57–68. doi:10.1016/j.micpro.2018.02.001.
[59] Hénon M. A two-dimensional mapping with a strange attractor. Commun Math Phys 1976;50(1):69–77. doi:10.1007/BF0160855.
[60] Shaoliang Y, Jiang T. Bifurcation and chaos in the Tinkerbell map. Int J Bifurc Chaos 2011;11:3137–56. doi:10.1142/S0218127411030581.
[61] Goldszztejn A, Hayes W, Collins P. Tinkerbell is Chaotic. Siam J Appl Dyn Syst 2011;10(4):1480–501.
[62] Chen L. An open-plus-closed-loop control for discrete chaos and hyperchaos. Phys Lett A 2001;281(5):327–33. doi:10.1016/S0375-9601(01)00055-X.
[63] Itoh M, Yang T, Chua LO. Conditions for impulsive synchronization of chaotic and hyperchaotic systems. Int J Bifurc Chaos 2001;11(2):551–60. doi:10.1142/S0218127401002262.
[64] May RM. Simple mathematical models with very complicated dynamics. Nature 1976;261:459–67.
[65] Verhulst PF. Recherches mathematiques sur la loi d' accroissement de la population. Nouv mém de l' Academie Royale des Sci et Belles-Lettres de Bruxelles 1845;18:1–41.
[66] Zhang Q, Guo L, Wei X. Image encryption using DNA addition combining with chaotic maps. Math Comput Model 2010;52(11–12):2028–35. doi:10.1016/j.mcm.2010.06.005.
[67] Zuras D, Cowlishaw M, Aiken A, Applegate M, Bailey D, Bass S, Bhandarkar D, Bhat M, Bindel D, Boldo S, et al. IEEE standard for floating-point arithmetic. IEEE Std 754-2008 2008:1–70. doi:10.1109/IEEESTD.2008.4610935.
[68] Bassham III Lawrence E, Rukhin AL, Soto J, Nechvatal JR, Smid ME, Barker EB, et al. SP 800-22 rev. 1a. a statistical test suite for random and pseudorandom number generators for cryptographic applications. Natl Inst Standards Technol 2010.
[69] Inzunza-González E, Cruz-Hernández C. Double hyperchaotic encryption for security in biometric systems. Nonlinear Dyn Syst Theory 2013;13(1):55–68.
[70] Fu C, Lin B, Miao Y, Liu X, Chen J-J. A novel chaos-based bit-level permutation scheme for digital image encryption. Optics Commun 2011;284:5415–23. doi:10.1016/j.optcom.2011.08.013.
[71] Gonzalo A, Li S. Some basic cryptographic requirements for chaos-based cryptosystems. Int J Bifurc Chaos 2006;16(8):2129–51. doi:10.1142/S0218127406015970.
[72] François M, Grosges T, Barchiesi D, Erra R. A new pseudo-random number generator based on two chaotic maps. Informatica 2013;24(2):181–97.
[73] François M, Grosges T, Barchiesi D, Erra R. Pseudo-random number generator based on mixing of three chaotic maps. Commun Nonlinear Sci NumerSimul 2014;19(4):887–95. doi:10.1016/j.cnsns.2013.08.032.
[74] Kerckhoffs A. La cryptographie militaire. Journal des sciences militaires 1883;IX:161–91.
[75] Murillo-Escobar MA, Cruz-Hernández C, Cardoza-Avendaño L, Méndez-Ramírez R. A novel pseudorandom number generator based on pseudorandomly enhanced logistic map. Nonlinear Dyn 2017;87(1):407–25. doi:10.1007/s11071-016-3051-3.
[76] Norouzi B, Mirzakuchaki S. A fast color image encryption algorithm based on hyper-chaotic systems. Nonlinear Dyn 2014. doi:10.1007/s11071-014-1492-0.
[77] Yuan Hong Mei LY. A new parallel image cryptosystem based on 5D hyper-chaotic system. Signal Process-Image Commun 2017;52:87–96. doi:10.1016/j.image.2017.01.002.
[78] Lee Wai Kong PR. SPRING: a novel parallel chaos-based image encryption scheme. Nonlinear Dyn 2018;92(2):575–93. doi:10.1007/s11071-018-4076-6.
[79] Wu Y, Noonan JP, Agaian S. NPCR and UACI randomness tests for image encryption. Cyber J: MultidiscipJSci Technol, J Sel Areas Telecommun(JSAT) 2011;1(2):31–8.